Ayushman Choudhuri

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Technical Skills

Programming: Python, C++, C

Frameworks & Libraries: ROS2, Pytorch, ONNX, Tensorflow, TensorRT, OpenCV, CUDA, Deepstreamer

MLOps: MLFlow, W&B, DVC, AWS Sagemaker, Terraform, Fiftyone

Developer Tools: Git, Docker, VS Code, Conda, CMake, Gazebo, RViz, Foxglove Studio, Vim, Tmux, Bash

Operating Systems: Linux

Languages: English (Native), German (B1)

Work Experience

Robotics Software Engineer - Computer Vision **December 2024 – Present**
Schmiede.one GmbH (Innovation lab of the Grimme group) *Düsseldorf, Germany*

- Built ROS2 object detection pipelines and custom vision algorithms for the Farmsort.one optical sorter, enabling real-time classification and sorting of agricultural produce.
- Architected the MLOps pipeline (auto-annotation, training, evaluation), cutting manual effort by 85% and accelerating model release cycles.
- Deployed and maintained production software and ML models, resolving customer issues to ensure stable, efficient system performance.

Master Thesis Student - Autonomous Driving **May 2024 – July 2025**
Institut für Kraftfahrzeuge (ika) RWTH Aachen University *Aachen, Germany*

- Researched an attention map–based methodology for explainable real-time 3D object detection using transformer models.
- Developed a ROS2 based pipeline to ingest LiDAR point clouds and display 3D object detection as well as saliency maps in real time.

Working Student - Computer Vision and Robotics **March 2024 – November 2024**
Schmiede.one GmbH (Innovation lab of the Grimme group) *Düsseldorf, Germany*

- Developed and field-validated a standalone harvest quality assurance system (QualiCam) at customer farms to assess crop quality in real-world conditions.
- Developed quantization and deployment pipelines for YOLO-based object detection models using TensorRT on NVIDIA Jetson Orin NX platform.

Intern - Computer Vision and Robotics **August 2023 – February 2024**
Schmiede.one GmbH (Innovation lab of the Grimme group) *Düsseldorf, Germany*

- Spearheaded the development of a stereo vision–based 3D object detection pipeline for an autonomous harvesters, enabling real-time collision avoidance. This project was showcased at Agritechnica 2023.
- Benchmarked and optimized state-of-the-art detection models on custom agricultural datasets, achieving 5× faster inference through ONNX and HailoRT quantization on Hailo8 edge processors.

Graduate Student Research Assistant **March 2023 – August 2023**
RWTH Aachen University *Aachen, Germany*

- Developed a perception pipeline for safe mobile robot operation in construction environments with a focus on stereo vision-based 3D object detection and LIDAR point cloud compression.
- Designed and deployed a closed-loop LIDAR tilt system to increase the vertical field of view by 100% for close-range applications. The ROS package was developed using C++ and deployed on an NVIDIA Jetson Xavier platform.

Software Developer - C++ **May 2020 – May 2021**
Synedyne Systems *Bangalore, India*

- Developed and implemented machine learning–based calibration algorithms on an edge device to estimate the payload of a self-loading cement mixer truck.
- Achieved a weight estimation accuracy of 98.5% with a maximum payload of 800 Kg.

Robotics Research Engineer **July 2019 – April 2020**
Indian Institute of Science *Bangalore, India*

- Designed and developed control software (C/C++) and a robotic test bed for precise liquid dispensing for composite manufacturing at the Department of Aerospace Engineering.
- Achieved an accuracy of up to +/- 10 microliters using a MEMS-based flow sensor.

Robotics Systems Engineer **July 2018 – July 2019**
Agilebot Automation *Bangalore, India*

- Worked on a AS/RS robotic system design for automating processes in medium and large scale warehouses.
- Developed the chassis and frame design for a pick and place autonomous rover module for medium scale warehouses.

Education

RWTH Aachen University **2021 - 2025**
MSc. Robotic Systems Engineering *Germany*

Manipal University **2014 - 2018**
B.Tech Mechanical Engineering *India*

Projects and Hackathons

- **mcap2hdf5** | Github Link
  Developing a tool to convert mcap files to hdf5 format for multimodal robot data.
- **Master Thesis**| Lab Link
  Explainable Transformer-based 3D Object Detection in LiDAR Point Cloud
- **TUM.ai Hackathon 2024** | Github Link
  Developed a multimodal AI system to detect and analyse deforestation from satellite images.
- **TUM.ai Hackathon 2023** | Github Link
  Developed an LLM-based AI consultant, tailored for the German Automotive Mittlestand.
- **Fraunhofer ICNAP Hackathon 2022**
  Developed a Time Series Classifier of Engine Data using Transformers. Won the 3rd place